

06 Matrices of animal populations

Although the paper focuses on a perennial plant data set, the same principles work for animal matrices too. Small sample sizes can compromise the estimation of survival and reproduction, especially if the analyst believes that matrix entries should vary by age, sex, or other environmental conditions. In some cases there are no available data, and the analysis proceeds using “guesses”, AKA prior beliefs. Using simple prior distributions for these parameters enables the calculation of credible intervals for both matrix entries and derived quantities in a natural way.

This example does not use the `raretrans` functions, because the information in Beissinger (1995) is provided as means and standard deviations, not as numbers of transitions.

```
library(tidyverse)
library(popbio)
library(huxtable)

# Raymond's theme modifications
rlt_theme <- theme(axis.title.y = element_text(colour="grey20",size=15,face="bold"),
  axis.text.x = element_text(colour="grey20",size=15, face="bold"),
  axis.text.y = element_text(colour="grey20",size=15,face="bold"),
  axis.title.x = element_text(colour="grey20",size=15,face="bold"))
```

Snail kites

Stephen Beissinger (1995) developed a PVA for the endangered snail kite in the Florida Everglades. A key piece of that problem was a periodic variation in water levels caused by management. Kites had lower survival and reproduction when water levels were drawn down in dry years to support irrigation. Data on radio telemetered animals and nest success were available for “high water” years, and anecdotal evidence was used to guesstimate values for “drought” and “lag” years. The model used three age classes because “[a] finer age-structured model was not feasible due to small sample sizes of age-specific demographic data”. Standard deviations and ranges were provided for some of the parameters. The methods describe survival values as drawn from a normal distribution.

```
b1995_table2 <- tribble(
  ~env_state, ~age_class, ~mean_ns, ~sd_ns, ~range_ns, ~prop_nesting, ~attempts, ~mean_s, ~sd_s, ~range_s,
  "", "", "Nesting success", "", "", "", "", "Annual survival", "", "",
  "Environmental state", "Age Class", "Mean", "SD", "Range", "Proportion Nesting",
  "Nesting attempts per year", "Mean", "SD", "Range",
  "Drought year", "Fledgling", "0.00", "0.00", "0.00-0.00", "0.00", "0.00", "0.50", "0.10", "0.30-0.80",
  "", "Subadult", "0.03", "0.10", "0.00-0.25", "0.15", "1.00", "0.60", "0.10", "0.40-0.90",
  "", "Adult", "0.03", "0.10", "0.00-0.25", "0.15", "1.00", "0.60", "0.10", "0.40-0.90",
  "Lag year", "Fledgling", "0.00", "0.00", "0.00-0.00", "0.00", "0.00", "0.85", "0.03", "0.75-0.92",
  "", "Subadult", "0.16", "0.10", "0.04-0.33", "0.15", "1.00", "0.90", "0.03", "0.80-0.95",
  "", "Adult", "0.16", "0.10", "0.04-0.33", "0.80", "1.50", "0.90", "0.03", "0.80-0.95",
  "High year", "Fledgling", "0.00", "0.00", "0.00-0.00", "0.00", "0.00", "0.90", "0.03", "0.75-0.92",
  "", "Subadult", "0.30", "0.10", "0.1-0.40", "0.25", "1.00", "0.95", "0.03", "0.85-0.98",
```

```

"", "Adult", "0.30", "0.10", "0.1-0.40", "1.00", "2.20", "0.95", "0.03", "0.85-0.98")

huxtable(b1995_table2) %>%
  set_colspan(row = 1, col = c(3,8), value = 3) %>%
  theme_article(header_col = FALSE) %>%
  set_width(0.9) %>%
  set_col_width(c(0.15, 0.1, 0.075, 0.075, 0.1, 0.15, 0.15, 0.075, 0.075, 0.1)) %>%
  set_position("left") %>%
  set_bottom_border(row = 1, col = everywhere, value = 0) %>%
  set_bottom_border(row = 2, col = everywhere, value = 1) %>%
  set_caption("Snail kite vital rates from Beissinger (1995), Table 2.")

```

Table 1: Snail kite vital rates from Beissinger (1995), Table 2.

Environmental state	Age Class	Nesting success			Proportion Nesting	Nesting attempts per year	Annual survival		
		Mean	SD	Range			Mean	SD	Range
Drought year	Fledgling	0.00	0.00	0.00-0.00	0.00	0.00	0.50	0.10	0.30-0.80
	Subadult	0.03	0.10	0.00-0.25	0.15	1.00	0.60	0.10	0.40-0.90
	Adult	0.03	0.10	0.00-0.25	0.15	1.00	0.60	0.10	0.40-0.90
Lag year	Fledgling	0.00	0.00	0.00-0.00	0.00	0.00	0.85	0.03	0.75-0.92
	Subadult	0.16	0.10	0.04-0.33	0.15	1.00	0.90	0.03	0.80-0.95
	Adult	0.16	0.10	0.04-0.33	0.80	1.50	0.90	0.03	0.80-0.95
High year	Fledgling	0.00	0.00	0.00-0.00	0.00	0.00	0.90	0.03	0.75-0.92
	Subadult	0.30	0.10	0.1-0.40	0.25	1.00	0.95	0.03	0.85-0.98
	Adult	0.30	0.10	0.1-0.40	1.00	2.20	0.95	0.03	0.85-0.98

Survival values

In stage structured matrices generally, the survival component of the matrix is a multinomial because of the possibility of transitioning between more than 2 stages. In an age structured matrix like this one, the only possibilities are survival to the next stage or death, so a simple $Beta(\alpha, \beta)$ prior is all that is needed. The mean is given by $\alpha / (\alpha + \beta)$, while the variance of the distribution is

$$\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

We have a mean and standard deviation from Table 1, and can obtain approximate α and β as follows

$$\nu = \frac{\mu(1 - \mu)}{\sigma^2}$$

$$\alpha = \nu\mu$$

$$\beta = (1 - \mu)\nu$$

as long as $\mu(1 - \mu) < \sigma^2$. This condition is met for all the values in Beissinger's Table 2.

```
kite_parms <- b1995_table2 %>%
  slice(-(1:2)) %>% # remove top two rows
  separate(col = range_ns, into = c("min_ns", "max_ns"), sep = "-") %>%
  separate(col = range_s, into = c("min_s", "max_s"), sep = "-") %>%
  mutate(across(.cols = 3:12, .fns = as.numeric)) %>% # convert to numeric
  mutate(nu_ns = mean_ns * (1 - mean_ns) / sd_ns^2,
         alpha_ns = mean_ns * nu_ns,
         beta_ns = (1 - mean_ns) * nu_ns,
         nu_s = mean_s * (1 - mean_s) / sd_s^2,
         alpha_s = mean_s * nu_s,
         beta_s = (1 - mean_s) * nu_s) %>%
  # set infinite values to missing
  mutate(across(where(is.numeric), ~ifelse(is.nan(.x), NA_real_, .x)))
kite_parms %>%
  select(env_state, age_class, alpha_ns, beta_ns, alpha_s, beta_s) %>%
  mutate(across(where(is.numeric), ~format(.x, digits = 2))) %>%
  bind_rows(tibble(env_state = c("", "Environmental State"),
                  age_class = c("", "Age Class"),
                  alpha_ns = c("Nest Success", "$\\alpha$"),
                  beta_ns = c("", "$\\beta$"),
                  alpha_s = c("Survival", "$\\alpha$"),
                  beta_s = c("", "$\\beta$")), .) %>%
  mutate(across(.cols = 3:6, ~case_when(grepl("NA", .x) ~ " ",
                                           TRUE ~ .x))) %>%
# filter(age_class != "Fledgling") %>%
hux() %>%
set_escape_contents(row = 2, col = 3:6, value = FALSE) %>%
theme_article(header_col = FALSE) %>%
set_na_string(value = " ") %>%
set_position("left") %>%
set_caption("Values of $\\alpha$ and $\\beta$ for each age class and
environmental state. Fledglings have zero probability of
nest success in all environmental states.")
```

One of the benefits of articulating the beliefs this way is that the reader can immediately see how much confidence the analyst has placed in each of the parameter estimates. The sum of α and β is effective sample size. So we have less confidence in the estimates of nest success in lag and drought years than high water years, as stated in Beissinger (1995). However, within each environmental state we have more confidence in the survival estimates of fledgling than subadult or adult stages, and less confidence in estimates from high

Table 2: Values of α and β for each age class and environmental state. Fledglings have zero probability of nest success in all environmental states.

Environmental State	Age Class	Nest Success		Survival	
		α	β	α	β
Drought year	Fledgling			12	12.5
	Subadult	0.087	2.8	14	9.6
	Adult	0.087	2.8	14	9.6
Lag year	Fledgling			120	21.3
	Subadult	2.150	11.3	90	10.0
	Adult	2.150	11.3	90	10.0
High year	Fledgling			90	10.0
	Subadult	6.300	14.7	50	2.6
	Adult	6.300	14.7	50	2.6

water years than lag years. Both of these are inconsistent with the prior beliefs stated by Beissinger(1995), but are a consequence of fixing the standard deviation of the estimates. Fixing the standard deviation gives a consistent belief about precision if the parameter is normally distributed, because the mean and standard deviation are independent. This is not true of a Beta distribution.

```
pdf_seq <- seq(-0.2,1.2,0.001)
plot_colors <- RColorBrewer::brewer.pal(3, "RdYlBu")
kite_parms2 <- kite_parms %>%
  mutate(env_state = rep(env_state[c(1,4,7)], each = 3),
         n_ns = alpha_ns + beta_ns,
         n_s = alpha_s + beta_s,
         lwr_ns = pbeta(min_ns, alpha_ns, beta_ns),
         upr_ns = pbeta(max_ns, alpha_ns, beta_ns),
         lwr_s = pbeta(min_s, alpha_s, beta_s),
         upr_s = pbeta(max_s, alpha_s, beta_s),
         plot_ns = map2(alpha_ns, beta_ns, ~tibble(x_ns = pdf_seq,
                                                  d_ns = dbeta(x_ns, .x, .y))),
         plot_ns_norm = map2(mean_ns, sd_ns, ~tibble(x_nsn = pdf_seq,
                                                  d_ns_norm = dnorm(x_nsn, .x, .y),
                                                  d_ns_tnorm = dnorm(x_nsn, .x, .y)/
                                                  pnorm(0, .x, .y, lower.tail = FALSE))),
         plot_s = map2(alpha_s, beta_s, ~tibble(x_s = pdf_seq,
                                                  d_s = dbeta(x_s, .x, .y))),
         plot_s_norm = map2(mean_s, sd_s, ~tibble(x_sn = pdf_seq,
                                                  d_s_norm = dnorm(x_sn, .x, .y),
                                                  d_s_tnorm = dnorm(x_sn, .x, .y)/
                                                  pnorm(1, .x, .y))),
  ) %>%
  unnest(cols = c("plot_ns", "plot_s", "plot_ns_norm", "plot_s_norm"),
        names_repair = "unique") %>%
```

```
mutate(env_state = factor(env_state, levels = c("Drought year", "Lag year", "High year")),
      age_class = factor(age_class, levels = c("Fledgling", "Subadult", "Adult")))
```

```
ggplot(data = kite_parms2) +
  geom_area(mapping = aes(x = x_s, y = d_s), alpha = 0.75, fill = plot_colors[1], color=plot_colors[1]) +
  geom_area(mapping = aes(x = x_sn, y = d_s_tnorm), alpha = 0.75, fill = plot_colors[3], color = plot_colors[3]) +
  facet_grid(env_state~age_class, scales = "free_x") +
  scale_x_continuous(limits = c(0,1)) +
  labs(x = "Annual survival", y = "Density") +
  theme_classic()
```

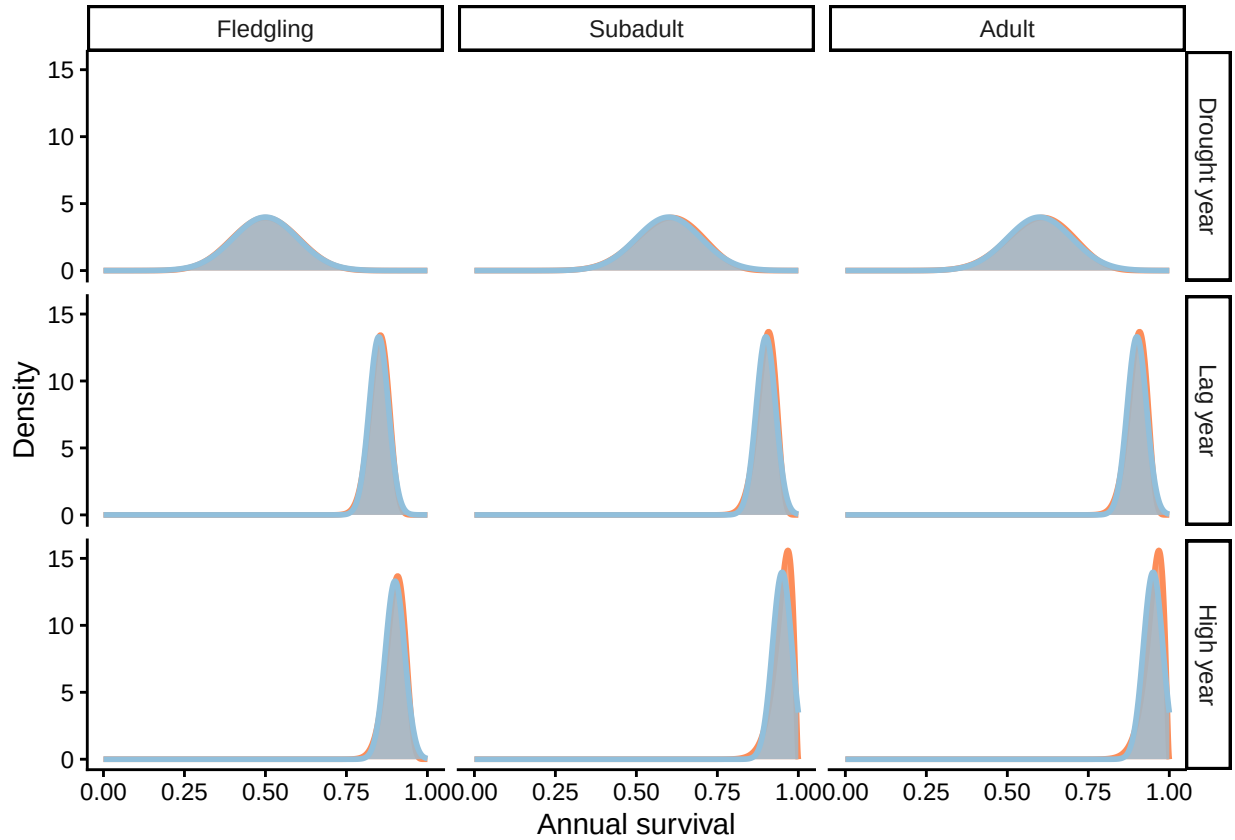


Figure 1: Probability distributions of annual survival by stage and environmental state. Beta distributions are red, truncated normal distributions are blue. Areas of overlap appear gray.

The differences in confidence are not easy to see in the density plots. The normal distribution is an excellent approximation to the beta distribution except for subadult and adult stages in high water years. Then the normal distribution has ~5% probability of values above 1. This was accounted for by rejecting inadmissible values and drawing another (Beissinger, Personal communication).

Reproductive rates

Beissinger (1995) only provides standard deviations for nest success, the probability of a nest fledging one or more chicks (Table 1). The reproductive rates of the matrices are the product of nest success, the number of young fledged per successful nest, the proportion of the population nesting, and the number of nest attempts

per year. Other than nest success all of these are treated as fixed values. A value for the number of fledglings per successful nest was not provided, but Snyder et al. (1989) had an average of ~ 2 per nest across all sites and years. Although this value could be used as is, the authors' Table 5 provides actual counts of young per successful nest, 55, 110, and 47 nests having 1, 2 or 3 young respectively. This knowledge is perfectly represented as a multinomial variable with a Dirichlet prior. Although the mean success does not vary across environmental states, the number of nests sampled does, and this should be reflected in the degree of confidence placed in each distribution. This representation also includes demographic stochasticity, so we will use a fixed value of 2 in what follows.

Beissinger (1995) used normal distributions for the uncertainty in nest success, but Table 3 in Snyder et al. (1989) provides values that can be converted into Beta priors for a binomial variable.

```
repro_parms <- tribble(
  # prior_young is vector of nests with 1, 2, or 3 young.
  # alpha_ns and beta_ns are the number of successful and unsuccessful nests
  ~env_state, ~age_class, ~prior_young, ~alpha_ns, ~beta_ns,
  "Drought year", "Subadult", c(1, 3, 1), 3, 91,
  "Drought year", "Adult", c(1, 3, 1), 3, 91,
  "Lag year", "Subadult", c(6, 25, 8), 11, 58,
  "Lag year", "Adult", c(6, 25, 8), 11, 58,
  "High year", "Subadult", c(48, 82, 38), 100, 236,
  "High year", "Adult", c(48, 82, 38), 100, 236
)

repro_parms2 <- repro_parms %>%
  mutate(n_ns = alpha_ns + beta_ns,
         plot_ns = map2(alpha_ns, beta_ns, ~tibble(x = pdf_seq,
                                                    d_ns = dbeta(x, .x, .y)))) %>%
  unnest(plot_ns) %>%
  mutate(env_state = factor(env_state, levels = c("Drought year", "Lag year", "High year")),
         age_class = factor(age_class, levels = c("Fledgling", "Subadult", "Adult")))

ggplot(data = filter(repro_parms2, age_class != "Fledgling"),
       mapping = aes(x = x, y = d_ns)) +
  geom_area(fill = plot_colors[2], alpha = 0.75, color=plot_colors[2], size = 1) + #Snyder etal 1989
  geom_area(data = filter(kite_parms2, age_class != "Fledgling"), #Beissinger 1995
           mapping = aes(x = x_ns, y = d_ns), fill = plot_colors[1], alpha = 0.75, color=plot_colors[1], size = 1) +
  geom_area(data = filter(kite_parms2, age_class != "Fledgling"), #Beissinger 1995
           mapping = aes(x = x_ns, y = d_ns_tnorm), fill = plot_colors[3], alpha = 0.75, color=plot_colors[3], size = 1) +
  facet_grid(env_state~age_class, scales = "free_y") +
  scale_x_continuous(limits = c(0,0.6)) +
  # scale_y_continuous(limits = c(0,30)) +
  labs(x = "Nest success", y = "Density") +
  rlt_theme
```

The confidence implied by the choices of the mean and variance in Beissinger (1995) is much less than the confidence implied by the sample sizes given in Snyder et al. (1989). It is possible that this is a deliberate choice because uncertainty in the other components of reproductive success was assumed to be zero.

Another advantage of using simple count priors for binomial and multinomial variables is that sampling from those variables automatically obeys restrictions on the numeric values. Beissinger (1995) sampled from normal distributions for survival and nest success. Many of these samples would have fallen outside the range [0,1] for a probability. Values outside this range were rejected and resampled (Beissinger, personal communication), giving “truncated” normal distributions. For nest success, this creates noticeable differences

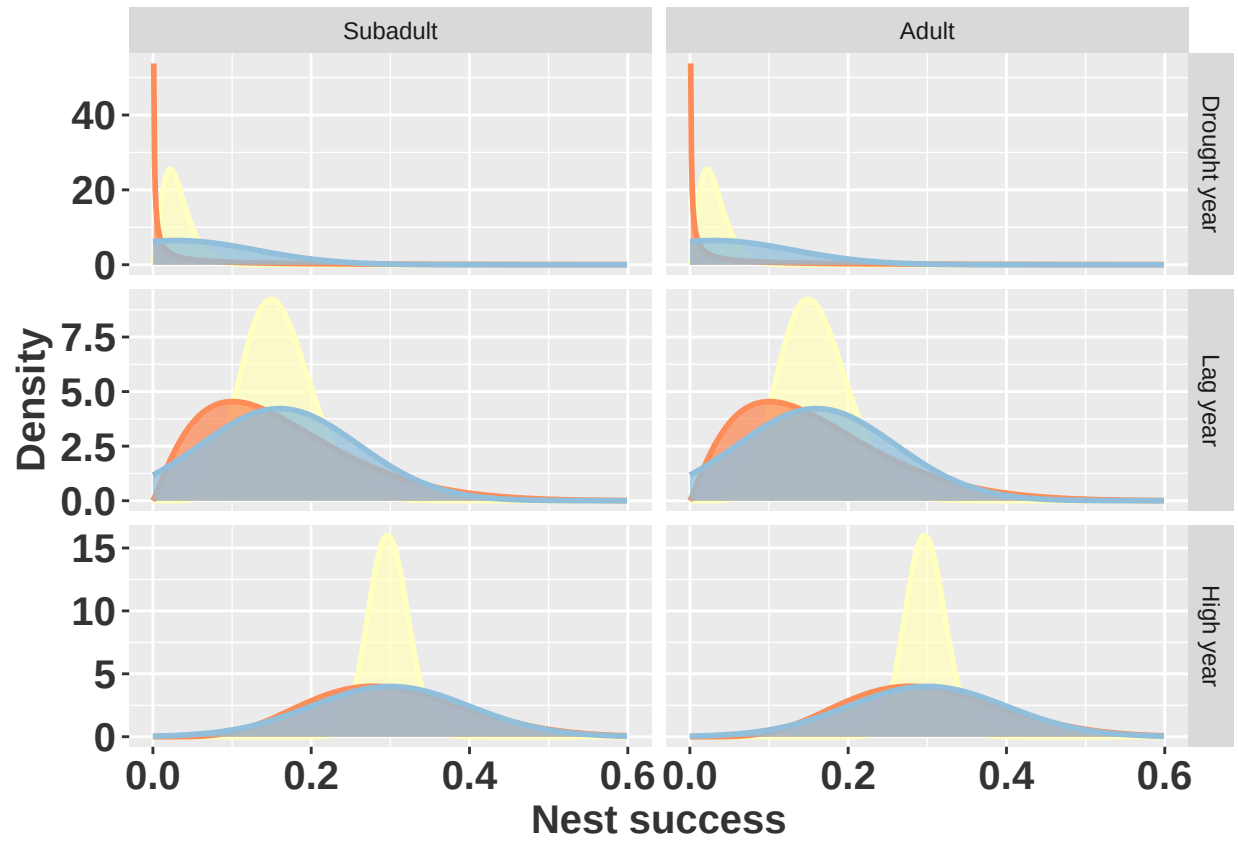


Figure 2: Prior beliefs about nest success using Beta distributions parameterized directly from Synder et al. (yellow, 1989), Beta distributions parameterized from Beissinger (red, 1995), and normal distributions from Beissinger (blue, 1995).

only in lag and drought years. In drought years particularly, the truncated normal distribution produces nest success values much higher than either of the parameterizations of the Beta distribution. The mean of the truncated normal is 0.092, 3 times larger than the intended mean.

Building the matrices

Now that we have distributions for the matrix entries, we can create a function that returns a matrix with values sampled from the relevant distributions.

```
# fix up parameter dataframes
surv_parms <- kite_parms %>%
  mutate(env_state = rep(env_state[c(1,4,7)], each = 3))
repro_parms <- repro_parms %>%
  left_join(select(surv_parms, env_state, age_class, prop_nesting, attempts))
snailkite_A <- function(e_state = "High year",
                        s_parms = surv_parms,
                        r_parms = repro_parms){
  # assumes young per successful nest fixed at 2
  A <- matrix(0, nrow = 3, ncol = 3)
  surv_parms <- filter(s_parms, env_state == e_state)
  repro_parms <- filter(r_parms, env_state == e_state)
  surv <- rbeta(3, surv_parms$alpha_s, surv_parms$beta_s)
  A[2,1] <- surv[2]
  A[3,2] <- A[3,3] <- surv[3]
  ns <- rbeta(2, repro_parms$alpha_ns, repro_parms$beta_ns)
  A[1,2:3] <- repro_parms$prop_nesting * repro_parms$attempts * ns * 2 * surv[1]
  A
}
snailkite_A()
#>      [,1] [,2] [,3]
#> [1,] 0.000 0.128 0.979
#> [2,] 0.973 0.000 0.000
#> [3,] 0.000 0.939 0.939
snailkite_A("Drought year")
#>      [,1] [,2] [,3]
#> [1,] 0.000 0.00791 0.0049
#> [2,] 0.644 0.00000 0.0000
#> [3,] 0.000 0.51006 0.5101
snailkite_A("Lag year")
#>      [,1] [,2] [,3]
#> [1,] 0.000 0.064 0.426
#> [2,] 0.895 0.000 0.000
#> [3,] 0.000 0.920 0.920
```

We can then draw many samples of each matrix to obtain credible intervals on derived quantities like the asymptotic growth rate λ .

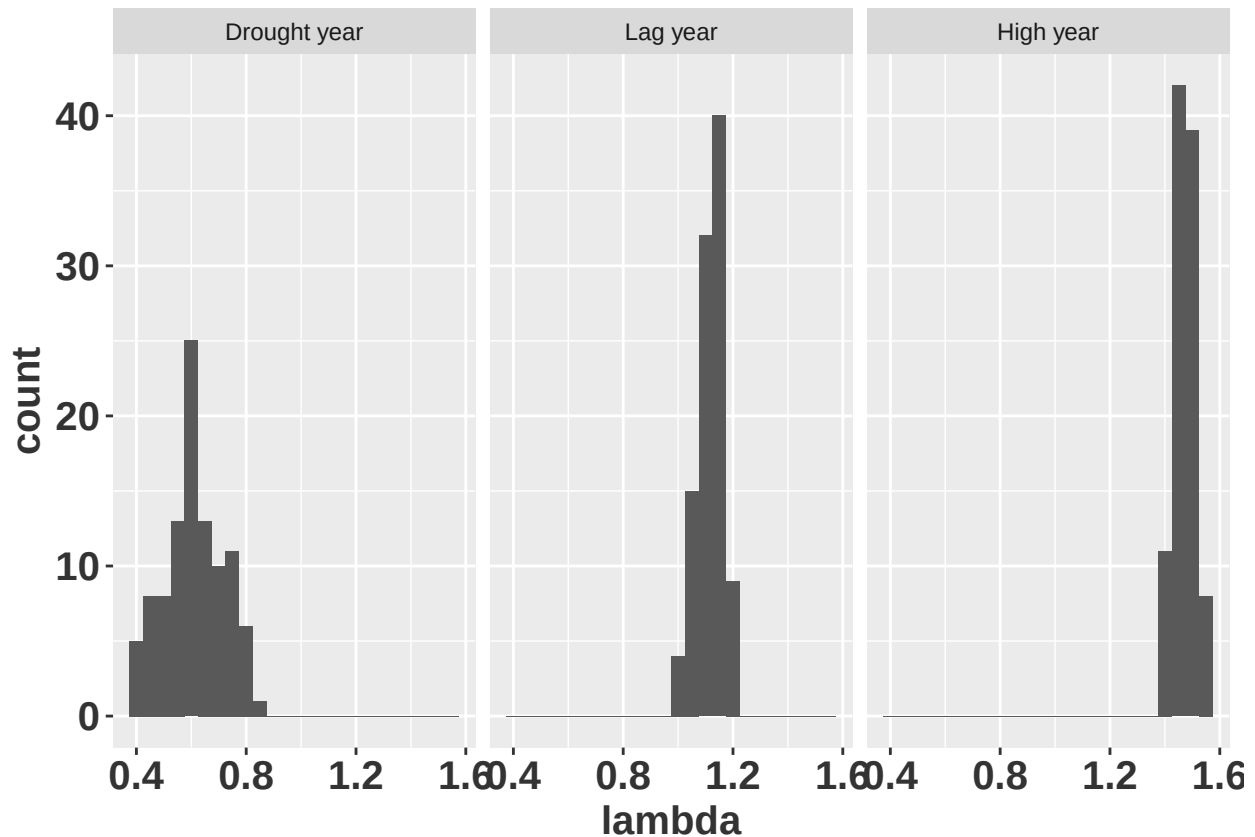
```
results <- tibble(
  e_state = factor(rep(c("Drought year", "Lag year", "High year"), each = 100),
    levels = c("Drought year", "Lag year", "High year")),
  A = map(e_state, snailkite_A),
  lambda = map_dbl(A, lambda)
```



```

)
ggplot(data = results,
       mapping = aes(x = lambda)) +
  geom_histogram(binwidth = 0.05) +
  facet_wrap(~e_state) +
  rlt_theme

```



We can reproduce extinction risk plots as in Beissinger (1995), if that is necessary.

Literature cited

Beissinger, Steven R. 1995. "Modeling Extinction in Periodic Environments: Everglades Water Levels and Snail Kite Population Viability." *Ecological Applications* 5 (3): 618–31. <http://www.jstor.org/stable/1941971>.

Snyder, Noel F. R., Steven R. Beissinger, and Roderick E. Chandler. 1989. "Reproduction and Demography of the Florida Everglade (Snail) Kite." *The Condor* 91 (2): 300–316. <http://www.jstor.org/stable/1368308>.